

MUHAMMAD FAREED

AI / ML Researcher & Engineer — efficient, deployable models

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SUMMARY

AI/ML researcher specializing in **efficient, deployable models**—parameter-efficient fine-tuning (QLoRA/LoRA), quantization, and small/multimodal models that run on modest hardware. MS Data Science **batch topper (CGPA 4.0/4.0)**; first-author **Q1** publication. Healthcare and low-resource-language AI as proving grounds; methods generalize across domains.

CORE SKILLS

Languages/Tools: Python, LaTeX, SQL, Google Colab, CUDA/cuDNN | **Frameworks:** PyTorch, TensorFlow, Hugging Face Transformers & PEFT, Keras

Efficient adaptation: QLoRA, LoRA, AdaLoRA, 4-bit quantization, knowledge distillation, on-premise & edge deployment |

Modelling: VLMs, LLMs, multimodal fusion, transfer learning, NLP, computer vision, generative AI

SELECTED PROJECTS

- **RU-SLM — Roman Urdu Small Language Model.** 4-bit QLoRA on an 8B model; ~74% fewer trainable parameters with deployable inference; reproducible pipeline.
- **TS-SLM — Text Summarization SLM.** LoRA + 4-bit quantization on Mistral-7B; length-controlled abstractive summarization; ROUGE/BERTScore and hallucination analysis.
- **Multilingual Sentiment & Language Detection.** PEFT on Gemma-3/Qwen-3; joint language + sentiment with structured JSON output; documented error modes.
- **CXRG-SVLM (MS thesis).** Small VLM generating chest X-ray reports on a single 16 GB GPU; 4-bit QLoRA (0.97% trainable); ROUGE-L 0.3116, clinical-entity F1 0.4867; 7.8 s/report.

PUBLICATIONS

- **Ethical Considerations of LLMs in Healthcare & Medicine** — *Frontiers in Digital Health* (2025), IF 3.8, Q1 — **first author, published.**
- **CXRG-SVLM: Parameter-Efficient VLM for Chest X-ray Report Generation** — *Scientific Reports* (2026) — **first author, under review.**
- **Optimizing mBERT for Punjabi Sentiment Analysis via PEFT** — *Computers, Materials & Continua* (2026) — **co-author, under review.**
- **LSTM-Based Deep Learning Models in Healthcare (systematic review)** — **first author, in preparation.**

EXPERIENCE

Science & Mathematics Teacher (BPS-14), Govt. Elementary School, Lahore — 2018–present. Permanent civil-service post held concurrently with MS, thesis, and four papers.

Digital Marketing & Web Development Trainer, Digiversity Training Institute, Lahore — 2022–2023.

Social Media Marketing Manager, Conversion Marketing Agency, Lahore — 2021–2022. Earlier: instructor/tutor roles in mathematics & physics, 2012–2017.

EDUCATION

MS in Data Science, RSCI, Riphah International University, Lahore — 2024–2026. CGPA 4.0/4.0; batch topper (first position, Spring 2024); full scholarship.

BS (Hons) in Mathematics, University of Education, Lahore — 2012–2016. First Division (67.45%).

AWARDS & CERTIFICATIONS

PMYP × Stanford Engineering CGOE AI cohort (2025) · Batch topper, MS Data Science (2024) · Dean's Honor List, Spring 2024 · Best Teaching Performance Award (2024). | **Certs:** Stanford CGOE; DeepLearning.AI Deep Learning Specialization; IBM GenAI/Transformers; Michigan Data Science Ethics.